

MINSK MAZOWIECKI, Poland

WMO: 123760

Lat: 52.1922N

Lon: 21.6635E

Elev: 184

StdP: 99.13

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-13.7	-20.1	0.6	-16.0	-16.9	0.9	-12.8	12.1	3.9	10.7	2.9	2.3	90	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.4	20.4	28.5	19.3	26.8	18.5	21.5	27.9	20.6	26.5	19.6	24.9	3.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.5	24.1	18.6	13.7	23.0	17.7	13.0	22.1	63.4	28.1	60.2	26.8	56.8	24.9	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.2	7.1	-20.8	33.7	4.5	1.3	-24.0	34.6	-26.7	35.4	-29.2	36.2	-32.5	37.1		
(5)			-20.1	23.2	4.6	1.1	-23.4	24.0	-26.1	24.7	-28.7	25.3	-32.1	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-3.2	-1.9	2.6	8.8	13.7	17.0	19.6	18.8	13.8	8.3	4.1	-0.1		
	DBStd	8.91	5.97	5.40	5.01	4.28	4.06	3.38	3.32	3.29	3.50	4.07	3.89	4.92		
	HDD10.0	1637	408	332	231	71	12	1	0	0	6	82	179	314		
	HDD18.3	3736	667	566	487	287	154	64	26	34	142	311	428	573		
	CDD10.0	1096	0	0	3	35	126	211	297	273	121	30	2	0		
	CDD18.3	153	0	0	0	1	9	24	64	48	6	0	0	0		
	CDH23.3	1464	0	0	0	15	107	252	579	452	59	0	0	0		
	CDH26.7	407	0	0	0	3	19	57	177	139	11	0	0	0		
(14) Wind	WSAvg	3.2	3.6	3.5	3.7	3.5	3.1	3.0	2.9	2.6	2.9	3.2	3.4	3.6		
(15) Precipitation	PrecAvg	582	34	31	34	39	58	69	82	65	53	38	40	38		
	PrecMax	771	96	68	71	99	135	150	280	210	141	119	110	75		
	PrecMin	439	4	5	10	3	22	18	21	7	12	4	2	9		
	PrecStd	72	19	13	15	20	26	34	52	37	35	26	19	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	10.5	17.6	25.3	29.0	31.0	32.7	33.3	28.3	21.2	14.9	10.5		
		MCWB	7.5	7.9	11.0	15.1	18.2	21.3	20.5	21.1	18.3	15.4	11.6	8.3		
	2%	DB	7.2	8.4	14.3	21.8	26.2	28.2	30.5	30.1	25.0	18.7	12.4	8.6		
		MCWB	5.6	6.1	8.7	13.2	17.4	19.5	20.1	20.7	16.8	14.0	10.0	7.3		
	5%	DB	5.7	6.5	12.1	19.2	23.8	26.2	28.6	27.9	22.6	16.5	10.5	7.1		
		MCWB	4.5	4.6	7.5	11.8	16.1	18.1	19.5	19.2	16.0	12.9	9.1	6.0		
	10%	DB	4.1	4.7	9.7	16.8	21.2	24.2	26.7	25.8	20.4	14.5	9.3	5.6		
		MCWB	3.1	3.3	6.5	10.3	14.5	17.1	18.8	18.3	14.9	12.0	8.1	4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	8.4	11.5	15.6	19.7	22.3	22.8	22.7	19.5	16.2	12.1	8.9		
		MCDB	8.5	9.9	15.9	23.6	26.4	29.0	29.4	29.6	26.0	20.2	14.4	9.9		
	2%	WB	5.9	6.3	9.6	14.0	18.1	20.7	21.6	21.3	17.8	14.6	10.6	7.5		
		MCDB	7.0	8.0	13.3	20.1	24.2	26.5	27.8	27.9	23.3	17.5	11.9	8.4		
	5%	WB	4.6	4.7	8.1	12.4	16.8	19.4	20.8	20.1	16.6	13.3	9.3	6.1		
		MCDB	5.6	6.4	11.1	17.9	22.5	24.5	26.6	26.2	21.0	15.8	10.3	6.9		
	10%	WB	3.1	3.4	6.6	10.9	15.6	18.0	19.9	19.1	15.6	12.2	8.2	4.8		
		MCDB	4.1	4.6	9.4	15.4	20.3	22.8	24.9	24.3	19.4	14.3	9.1	5.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.8	7.9	10.6	10.8	11.2	11.0	11.1	9.8	7.9	4.9	4.4		
		MCDBR	4.9	7.3	11.7	14.5	13.9	13.8	13.8	14.2	13.4	10.9	7.1	5.4		
	5% WB	MCWBR	4.2	5.4	7.1	7.5	6.7	6.6	5.5	6.1	6.4	6.6	5.3	4.5		
		MCWBR	4.7	6.8	10.7	13.6	12.8	12.3	12.0	12.5	11.7	9.8	6.4	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.342	0.375	0.428	0.412	0.409	0.439	0.433	0.394	0.365	0.354	0.312		
	taud		2.225	2.215	2.219	2.134	2.241	2.280	2.212	2.223	2.302	2.358	2.282	2.269		
	Ebn at Noon		661	754	805	803	840	846	810	791	779	726	608	598		
	Edh at Noon		68	92	112	138	130	126	133	125	103	80	66	56		
(44) All-Sky Solar Radiation	RadAvg		0.70	1.42	2.64	4.08	5.19	5.50	5.13	4.50	3.18	1.77	0.74	0.49		
	RadStd		0.14	0.23	0.36	0.55	0.52	0.47	0.61	0.37	0.48	0.29	0.07	0.07		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (21 neighbors)		+0.50	N/A	N/A	N/A	+0.41	N/A	N/A	-165	+80	+28	

Nomenclature: See separate page